

Towards a New Dynamic Influence Model

Some Ideas on Influence Redistribution Driven by Homophily

Christian Vargas

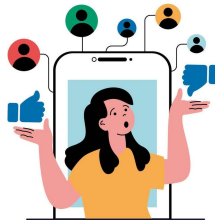
`vargas.christian@correounivalle.edu.co`

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- In social networks, influence is **not static**: the impact individuals have on one another evolves over time.
- **Homophily** is a social phenomenon in which individuals tend to interact more with others who are similar to them.
- In opinion dynamics models, similarity can be measured through differences in agents' opinion states.

- Core idea: introduce **dynamic influence weights** in the network.
- At each iteration, **influence is redistributed** as a function of opinion similarity.
- In this sense, agents reallocate influence over their neighbors based on perceived homophily.



2. Dynamic Influence Update Process

- Let $B_i(t)$ denote the opinion of agent i at time t .
- Define a distribution function $f : [0, 1] \rightarrow (0, \infty)$.
- The quantity $\Delta_{ij}^t = |B_i(t) - B_j(t)|$ represents the level of (indirect) homophily between agents i and j at time t .
- Influence is redistributed according to

$$w_{ij}(t+1) = \frac{f(\Delta_{ij}^t)}{\sum f(\Delta_{ik}^t)}, \quad \sum f(\Delta_{ij}^t) \neq 0.$$

2. Dynamic Influence Update Process

Requirements that the function f should satisfy:

- f should attain its maximum at $\Delta = 0$.
Agents with identical opinions receive the highest influence weight.
- f should be monotonically decreasing in Δ .
Influence decreases as opinion difference increases.
- The shape of f should encode the willingness of individuals (or society) to consider differing opinions.

$$f(0) = \max_{\Delta \in [0,1]} f(\Delta) \quad f'(\Delta) \leq 0, \quad \Delta \in [0,1]$$

3. Candidate Functions For f

Some authors [1, 2, 3] have proposed influence update rules based on functions similar to f (f_1 , f_2 , and f_3 , respectively).

1		$f_1(x) = e^{-c \cdot x} \{0 \leq x \leq 1\}$
2		$f_2(x) = e^{-c \cdot x^2} \{0 \leq x \leq 1\}$
3		$f_3(x) = \frac{1}{1 + c \cdot x} \{0 \leq x \leq 1\}$
4		$f_4(x) = \frac{1}{1 + c \cdot x^2} \{0 \leq x \leq 1\}$

These functions satisfy the structural properties required for f to serve as a valid influence distribution mechanism.

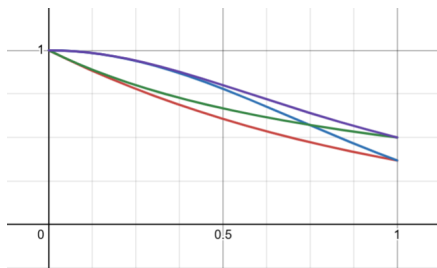


Figure 1: f_1 , f_2 , f_3 and f_4 for $\Delta_{ij}^t \in [0, 1]$ when $c = 1$.

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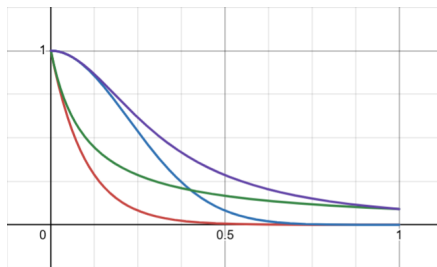


Figure 2: f_1 , f_2 , f_3 and f_4 for $\Delta_{ij}^t \in [0, 1]$ when $c = 10$.

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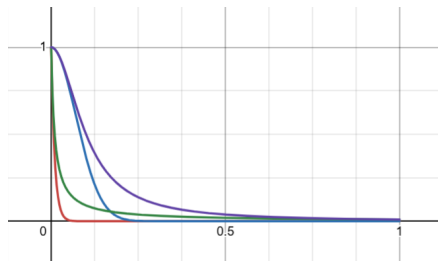


Figure 3: f_1 , f_2 , f_3 and f_4 for $\Delta_{ij}^t \in [0, 1]$ when $c = 100$.

4. Some Simulations

Let's consider a polarized network with weak cross-group influence, and analyze whether homophily-driven updates can potentially prevent consensus.

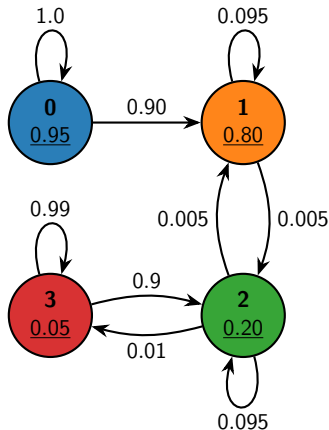


Figure 4: A polarized network.

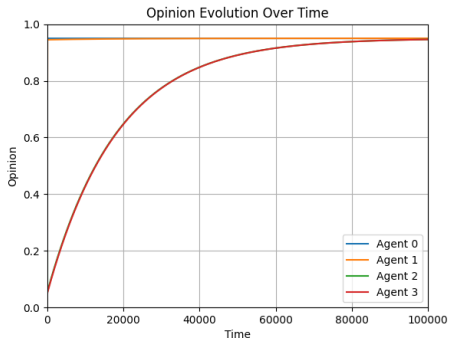
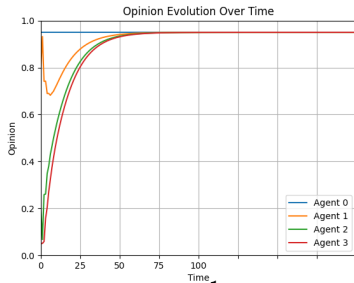
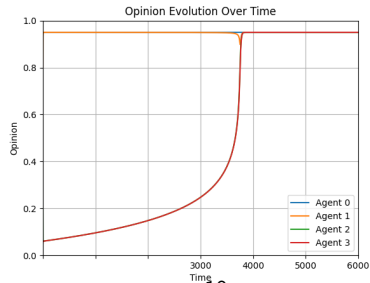


Figure 5: Opinion evolution under static influence.

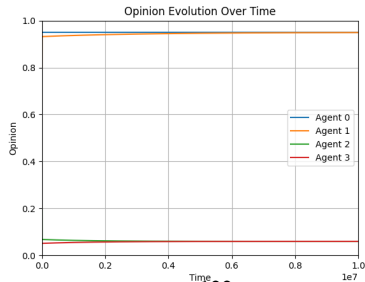
4. Simulations with $f_1(\Delta_{ij}) = e^{-c} \Delta_{ij}$



$c = 1$



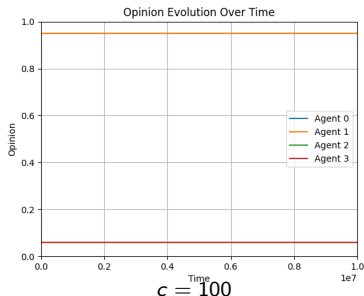
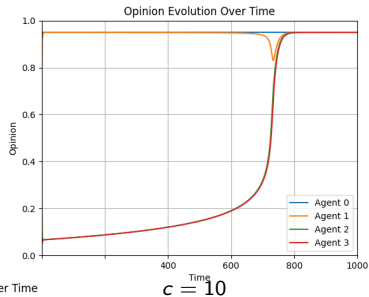
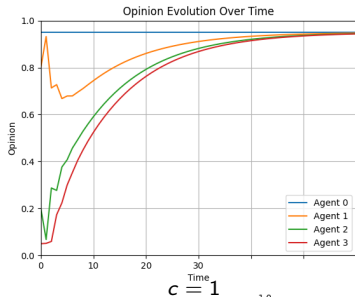
$c = 10$



$c = 100$

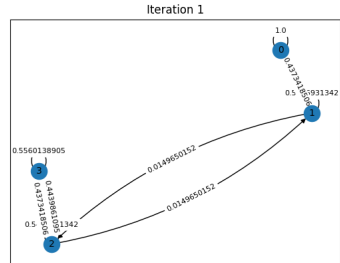
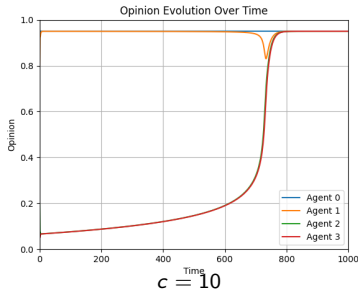
For large c , cross-group influence becomes negligible, and consensus appears unreachable even after 10^7 iterations.

4. Simulations with $f_2(\Delta_{ij}) = e^{-c} \Delta_{ij}^2$

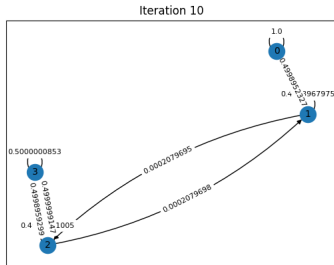


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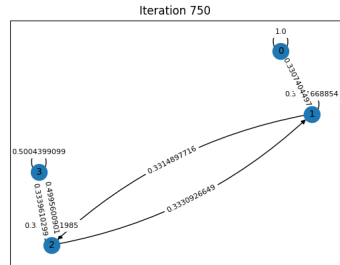
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Influence graph after 1 iteration.

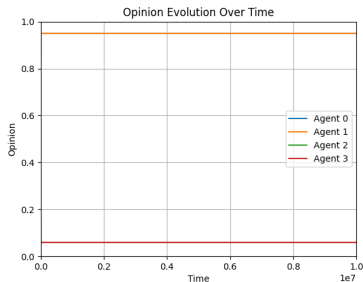


Influence graph after 10 iteration.



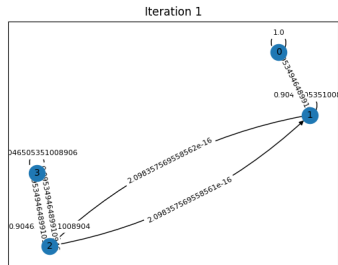
Influence graph after 750 iteration.

4. Simulations with $f_2(\Delta_{ij}) = e^{-c} \Delta_{ij}^2$

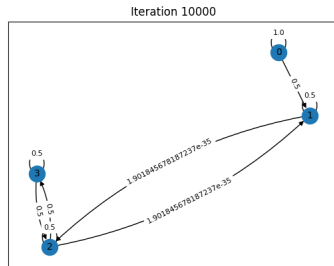


$c = 100$

It is not only the reduction of influence that matters, but the speed at which these connections vanish toward zero.



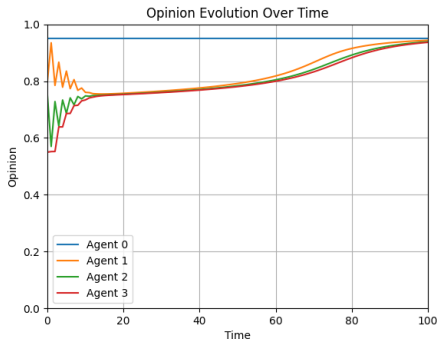
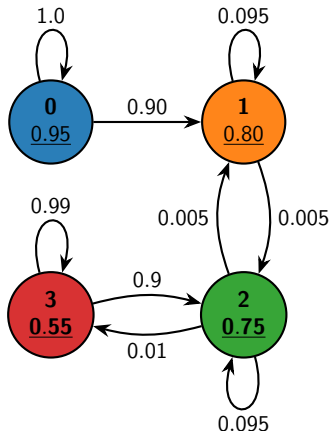
Influence graph after 1 iteration.



Influence graph after 10 000 iterations.

4. Simulations with $f_2(\Delta_{ij}) = e^{-c} \Delta_{ij}^2$

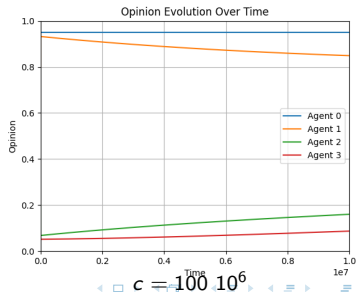
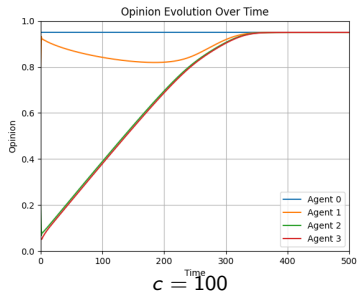
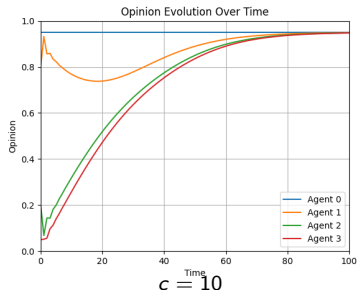
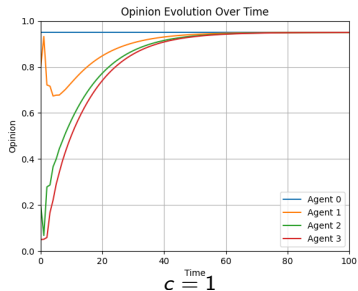
Let's consider the same network topology but with a different initial opinion state, one that is less polarized.



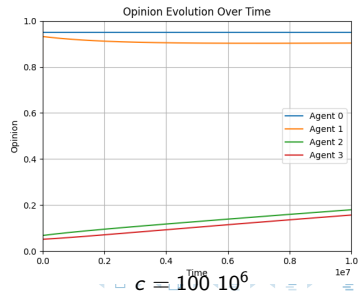
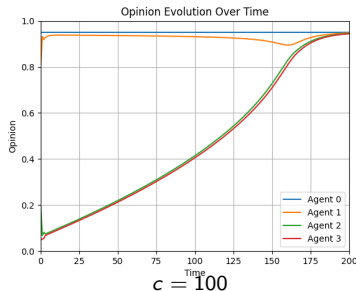
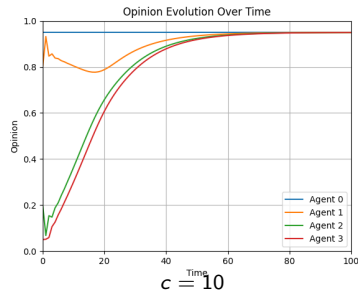
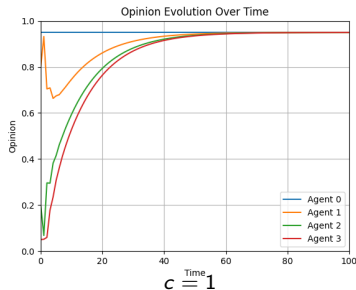
$c = 100$

Figure 6: Same topology with closer initial opinions.

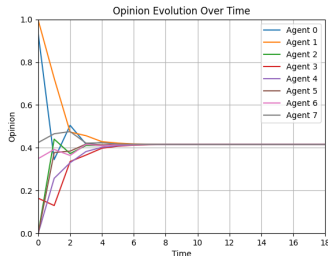
4. Simulations with $f_3(\Delta_{ij}) = \frac{1}{1 + c \Delta_{ij}}$



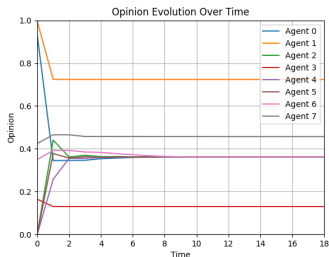
4. Simulations with $f_4(\Delta_{ij}) = \frac{1}{1 + c \Delta_{ij}^2}$



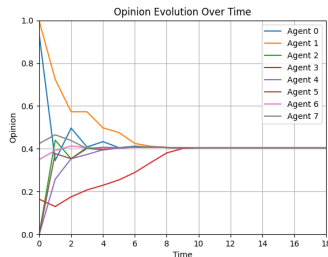
4. Simulations on a Non-Polarized Network



Network that converges rapidly to consensus.

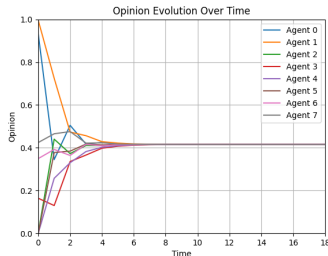


$f_1(\Delta_{ij}), c = 100$, diverges.

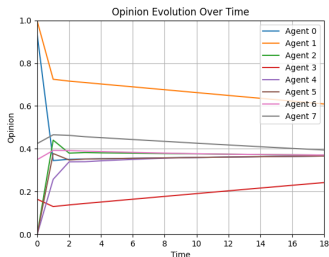


$f_2(\Delta_{ij}), c = 100$, converges.

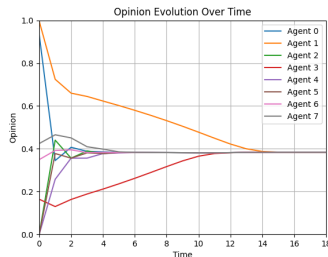
4. Simulations on a Non-Polarized Network



Network that converges rapidly to consensus.

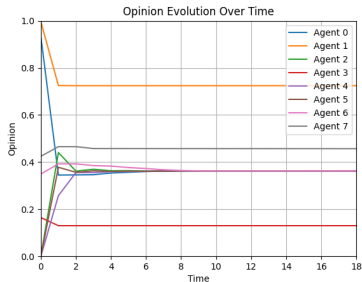


$f_3(\Delta_{ij}), c = 1000$, converges.



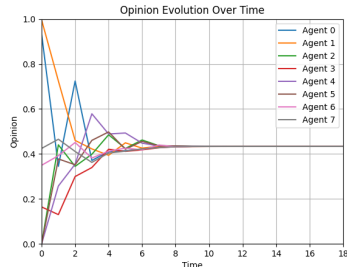
$f_4(\Delta_{ij}), c = 1000$, converges.

4. Simulations on a Non-Polarized Network

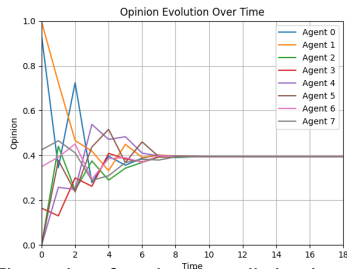


$f_1(\Delta_{ij}), c = 100$, diverges.

When the update function is highly selective, excluding self-influence can prevent early fragmentation and promote convergence.



The agent's self-influence remains static.



The update function is applied only when

$$0.01 < \Delta_{ii} < 0.99.$$

- A **homophily-driven redistribution rule** is introduced to **update influence weights** in a **fixed** social network (no new connections are created).
- The redistribution function $f(\Delta_{ij}^t)$ may take multiple forms; its shape determines how influence decays with opinion distance, thereby **encoding the willingness** of agents (or society) **to consider differing opinions**.
- Smooth, **bell-shaped functions** (e.g., Gaussian-type) are particularly well suited: they prioritize similar opinions while preserving limited openness to disagreement.

- Function **f_4 remains unexplored** in the literature. It is well suited for modeling societies that exhibit homophily without completely isolating opposing views; however, it is less appropriate for highly polarized settings. In contrast, **f_2 appears more versatile** across different polarization regimes.
- **A key design decision concerns self-influence:** including it in the redistribution rule systematically prioritizes an agent's own opinion, whereas keeping it fixed preserves a constant level of self-trust throughout the simulation.

- [1] Miao He and Xiaoming John Zhang. Affinity, value homophily, and opinion dynamics: The coevolution between affinity and opinion. *PLOS ONE*, 18, 11 2023.
- [2] Alan Tsang and Kate Larson. Opinion dynamics of skeptical agents. In *Proceedings of the 2014 International Conference on Autonomous Agents and Multi-Agent Systems*, pages 277–284. International Foundation for Autonomous Agents and Multiagent Systems, 2014.
- [3] Arthur Carvalho and Kate Larson. A consensual linear opinion pool. In *Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence*, pages 2518–2524. AAAI Press, 2013.

Thank you for your attention

For questions or comments, please contact:

`vargas.christian@correounivalle.edu.co`